Video-Based Digital Loan
Origination & Risk Assessment

LOANVISION AI

OUR TEAM

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Hackathon — Problem Statement 3

PROBLEM PREVIEW

Traditional digital loan journeys face high drop-offs, fraud risks, incomplete data capture, and limited contextual understanding of customers. Manual KYC adds operational overhead while form-based applications lack real-time verification and intent validation.

KEY CHALLENGES

60%+	Application drop-off rate
3–7 days	Avg. loan turnaround time
₹12K+	Cost per manual KYC
35%	Fraud detected post-disbursal

SOLUTION OVERVIEW

A single video call replaces the entire loan application journey — from KYC to offer generation.

Video-First Onboarding

Live WebRTC call captures identity, consent, and data in a single interaction

Agentic AI Orchestrator

LLM agent drives conversation, extracts entities, classifies risk in real-time

Multi-Layer Verification

Face match, age estimation, geo-fencing, and liveness detection prevent fraud

Instant Offer Generation

Policy engine + ML scoring produce personalized offers before call ends

CUSTOMER JOURNEY

01

Campaign Link

Customer receives secure link via SMS, WhatsApp, or email campaign

02

Video Call

WebRTC session starts with geo-location and device metadata capture

03

AI Interview

LLM agent conducts structured conversation; STT extracts employment, income, consent

04

Identity Check

Face match against ID, liveness challenge, age estimation via computer vision

05

Risk Scoring

Bureau data + ML propensity model + fraud signals produce composite score

06

Offer Delivered

Personalized loan offer with amounts, rates, tenure, and EMI shown in-call

SYSTEM ARCHITECTURE

PRESENTATION LAYER

WebRTC Video Client · Campaign Landing Page · Mobile SDK · Responsive UI

AI & PROCESSING LAYER

Whisper STT · LLM Agent (Claude) · CV Pipeline (Face+Age) · NER & Intent

DECISION & POLICY LAYER

Rule Engine (OPA) · ML Risk Scorer · Bureau API Gateway · Offer Generator

DATA & COMPLIANCE LAYER

Encrypted Vault · Audit Logger · Consent Store · CKYC / AA Integration

CORE AI MODULES

SPEECH-TO-TEXT ENGINE

OpenAI Whisper large-v3 for multilingual STT
Real-time streaming with Voice Activity Detection
Hindi + English code-switching support
Automated punctuation and speaker diarization

COMPUTER VISION PIPELINE

MTCNN face detection + ArcFace embedding match
Age estimation via DEX (Deep Expectation) CNN
Anti-spoofing liveness: blink, head-turn challenges
Geo-tagged frame capture for audit trail

LLM AGENTIC ORCHESTRATOR

Claude / GPT-4 as the conversational backbone
Tool-calling: triggers bureau lookup, policy check, offer gen
Entity extraction: employer, income, purpose, consent
Structured JSON output for downstream decisioning

RISK & FRAUD DETECTION

Multi-Signal Detection

Geo-Fence Mismatch

GPS vs. declared address deviation > 50 km

Age Inconsistency

CV-estimated vs. ID-declared age differs > 5 years

Liveness Failure

Anti-spoofing challenge not passed (blink/turn)

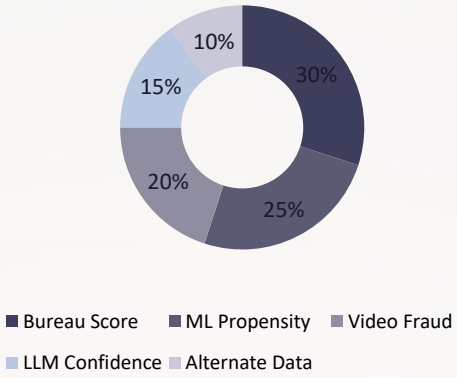
Voice Stress Analysis

Pitch variance and hesitation on key questions

Device Fingerprint

Known fraud device / VPN / emulator detection

Composite Risk Score



Weighted ensemble of rules + ML + LLM confidence

OFFER GENERATION

SAMPLE OFFER CARD

Eligible Amount	**₹ 5,00,000**
Interest Rate	**10.5% p.a.**
Tenure Options	**12 / 24 / 36 months**
EMI (36 mo.)	**₹ 16,267 / month**
Processing Fee	**1.5% (₹ 7,500)**
Risk Band	**A2 — Low Risk**

OFFER ENGINE LOGIC

1. Policy filter — eligibility rules, product match, geo restrictions
2. Bureau pull — CIBIL/Experian score, existing obligations, DPD history
3. ML scoring — propensity model + PD/LGD estimation for pricing
4. LLM enrichment — classify purpose, assess confidence, flag anomalies
5. Offer assembly — eligible amounts, rates, tenures, EMI calculations
6. Approval routing — auto-approve up to threshold, else escalate

TECHNOLOGY STACK

Video & Media	WebRTC · MediaSoup SFU · FFmpeg · Canvas API
Speech & NLP	Whisper large-v3 · Indic-BERT · spaCy NER · LangChain
Computer Vision	MTCNN · ArcFace · DEX Age Model · OpenCV + ONNX
LLM & Agents	Claude Sonnet 4 via API · Tool-calling · Structured Output
Backend	FastAPI (Python) · Node.js (signaling) · Redis · Kafka
Infrastructure	Kubernetes · AWS EKS · S3 + KMS · Prometheus + Grafana
Data & Compliance	PostgreSQL · MongoDB · AES-256 at rest · mTLS in transit

SECURITY & COMPLIANCE

DATA PROTECTION

- AES-256 encryption at rest for all PII & video recordings
- mTLS for all inter-service communication
- PII tokenization — raw data never leaves the vault
- Role-based access control (RBAC) with audit logging
- Auto-expiry of video recordings per retention policy

REGULATORY ALIGNMENT

- RBI Video KYC guidelines (2020 & 2022 updates)
- CKYC integration for central identity registry
- Account Aggregator framework for consent-based data
- DPDP Act 2023 — purpose limitation, consent mgmt
- Full audit trail: video, transcript, decision rationale

SCALABILITY & RESULTS

500+	<300ms	<2s	99.9%
Concurrent Video Calls	STT Latency	Risk Score Computation	Uptime SLA

ARCHITECTURE DECISIONS

- **Microservices on K8s**
  Each AI module as an independent, auto-scaling pod
- **Event-Driven (Kafka)**
  Decoupled processing for video, audio, and decisions
- **GPU Pool for Inference**
  Shared GPU nodes with request batching for efficiency
- **Edge CDN + TURN**
  Static assets via CloudFront; geo-distributed TURN servers

IMPLEMENTATION ROADMAP

01 **Week 1–3** **Foundation**

WebRTC video pipeline · STT integration · Basic LLM agent · Campaign link flow

02 **Week 4–6** **Intelligence**

Face match & liveness · Age estimation · NER entity extraction · Bureau API

03 **Week 7–9** **Decisioning**

Risk scoring ensemble · Policy rule engine · Offer generation · Auto-approval

04 **Week 10–12** **Production**

Security hardening · Load testing · Compliance audit · Monitoring · Go-live

WHY WE WIN

Agentic, Not Scripted

LLM agent adapts conversation dynamically — handles edge cases, regional languages, and follow-ups without rigid scripting

Multi-Modal Fraud Detection

Combines face, voice, location, device, and behavioural signals — not just document verification

Deterministic + Probabilistic

LLM enriches but never overrides policy rules. Hard guardrails on eligibility; AI adds intelligence on top

Sub-8-Minute Onboarding

From campaign click to offer delivery in a single video call — 80% reduction vs traditional journey

Revenue Impact

Higher conversion (fewer drop-offs), lower fraud losses, reduced operational cost per application